
Co-Stimulatory Signal Architecture as a Primary Control Layer in Engineered Natural Killer Cell Immunotherapy

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Abstract

Natural killer (NK) cells possess intrinsic cytotoxic capacity against malignant cells, yet their clinical efficacy has been constrained by insufficient activation within the tumor microenvironment, limited persistence, and susceptibility to immunosuppression. Existing NK-based therapeutic strategies primarily address downstream deficiencies—most notably through cytokine supplementation or checkpoint inhibition—without directly resolving the upstream constraints governing activation commitment.

This paper advances a mechanistic framework in which **co-stimulatory signal architecture constitutes the primary control layer of NK cell function**, determining activation threshold, signal integration, and durability of effector responses. We propose that engineered NK cells can be reprogrammed through **synthetic intracellular adapter integration**, coupling NK-native activating receptors to defined signaling modules that enhance signal gain and persistence while preserving antigen-independent tumor recognition.

Within this hierarchy, **interleukin-15 (IL-15)** is repositioned as a stabilizing factor that maintains activated states, and **attenuation of inhibitory pathways** is defined as a protective mechanism that preserves signaling integrity rather than initiating activation. We further outline candidate molecular architectures, predicted intracellular signaling outcomes, and safety considerations associated with constitutive co-stimulatory engineering.

This framework reframes NK immunotherapy as a problem of **intracellular signal design**, enabling durable, systemic, and potentially antigen-agnostic cancer control.

1. Introduction

Natural killer cells are key effectors of the innate immune system, capable of eliminating transformed or stressed cells without prior antigen sensitization. Their activity is governed by the integration of activating and inhibitory signals, enabling discrimination between healthy and abnormal cells.

Despite this intrinsic capability, NK cell-based immunotherapies have not yet achieved the clinical impact observed with T cell-based approaches. Limitations include:

- Weak or transient activation in tumor environments
- Rapid functional exhaustion or contraction
- Suppression by tumor-derived inhibitory signals

Current therapeutic strategies—such as cytokine administration, checkpoint blockade, and chimeric antigen receptor (CAR) engineering—have improved NK activity but do not fully address the **core problem of activation instability**.

We propose that the dominant limitation in NK immunotherapy is not recognition, but **insufficient intracellular signal architecture to sustain activation once initiated**.

2. NK Cell Signal Integration: Native Architecture

NK activation is mediated by a distributed receptor network:

- Activating receptors: NKG2D, NKp30, NKp44, NKp46
- Co-activation receptors: DNAM-1 (CD226), 2B4 (CD244)
- Fc receptor: CD16 (antibody-dependent cytotoxicity)

These receptors signal through intracellular adapters including:

- DAP10 → PI3K/AKT signaling
- DAP12 → ITAM-mediated Syk/ZAP70 activation
- CD3ζ and FcRγ → ITAM-based signal amplification

Inhibitory receptors (e.g., KIRs, NKG2A) counterbalance activation via SHP-1/2 phosphatases.

This system lacks a dominant co-stimulatory axis analogous to CD28 in T cells. Instead, NK cells rely on **cumulative signal integration**, making them vulnerable to:

- Reduced ligand density
 - Spatial fragmentation of activating signals
 - Increased inhibitory tone
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3. Co-Stimulatory Signal Architecture as the Primary Control Layer

3.1 Activation Threshold and Signal Collapse

Tumors rarely eliminate all activating ligands. Instead, they:

- Lower ligand expression
- Disrupt synapse formation
- Increase inhibitory signaling

This results in **sub-threshold activation**, where NK cells engage targets but fail to commit to cytotoxic programs.

3.2 Synthetic Co-Stimulatory Integration

To overcome this, engineered NK cells can incorporate **synthetic co-stimulatory signaling modules** integrated into NK-native receptor pathways.

Design Principle

Rather than replacing recognition (as in CAR-T), synthetic domains are **coupled to endogenous receptors** to amplify natural tumor sensing.

Candidate Architectures

1. **DAP10–CD28 Hybrid**
 - Enhances PI3K/AKT signaling
 - Lowers activation threshold
 - Increases sensitivity to low ligand density
 2. **DAP12–4-1BB Hybrid**
 - Combines ITAM signaling with NF- κ B activation
 - Promotes mitochondrial fitness and persistence
 3. **CD3 ζ Augmentation of CD16 Pathway**
 - Amplifies antibody-dependent cytotoxicity
 - Increases signal strength during immune complex engagement
 4. **2B4 (CD244)–based Co-stimulation Tuning**
 - Modulates SAP-dependent signaling
 - Balances activation vs exhaustion
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4. Intracellular Signaling Effects

Engineered co-stimulatory architectures are predicted to:

- Increase PI3K/AKT pathway activation → lower activation threshold
- Enhance NF-κB signaling → sustained transcriptional activation
- Improve mitochondrial biogenesis → metabolic fitness
- Stabilize immune synapse formation → prolonged target engagement

This converts weak, transient signals into **stable activation states**.

5. IL-15 as a Persistence Stabilizer

5.1 Role of IL-15

Interleukin-15 is the dominant cytokine supporting NK survival, proliferation, and metabolic competence.

5.2 Functional Reframing

In this architecture:

- Co-stimulation → initiates activation
- IL-15 → preserves and extends that state

5.3 Engineering Strategies

- Membrane-bound IL-15/IL-15Rα complexes
- Controlled autocrine IL-15 expression
- Cis-presentation to maintain localized signaling

5.4 Effects

- Prevents post-activation contraction
 - Sustains cytotoxic granule production
 - Supports long-term persistence
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6. Resistance to Immunosuppression

Tumor environments suppress NK cells through:

- TGF-β signaling

- Adenosine pathways
- Checkpoint receptors (e.g., TIGIT, NKG2A)

Engineering Approaches

- Dominant-negative TGF- β receptors
- Partial attenuation of inhibitory receptors
- Intracellular signaling insulation

Role in Architecture

Suppression resistance:

- Does not initiate activation
 - Preserves already established signaling states
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7. Risk Analysis and Safety Considerations

7.1 Tonic Signaling

Constitutive co-stimulation may cause:

- Chronic activation
- Cellular exhaustion
- Reduced lifespan

7.2 Off-Target Cytotoxicity

Lower activation thresholds may:

- Increase killing of stressed but healthy cells

7.3 Cytokine Dysregulation

Autocrine IL-15 may lead to:

- Excessive proliferation
- Systemic inflammation

7.4 Signal Imbalance

Overweighting activating pathways may:

- Override inhibitory safeguards
 - Disrupt immune homeostasis
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8. Design Constraints for Clinical Translation

Effective implementation requires:

- Controlled promoter systems (e.g., EF1 α -class for moderate expression)
 - Stable genomic integration (safe-harbor loci)
 - Functional assays measuring activation durability
 - Built-in safety switches (e.g., inducible apoptosis systems)
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9. Implications for Antigen-Agnostic Therapy

Unlike CAR-based strategies, this framework:

- Preserves broad stress-based tumor recognition
- Reduces dependence on single antigens
- Limits immune escape through antigen loss

This enables **systemic, non-invasive cancer immunotherapy** capable of targeting heterogeneous tumors.

10. Conclusion

NK cell immunotherapy is fundamentally constrained by insufficient activation stability rather than recognition failure. By engineering **co-stimulatory signal architecture as the primary control layer**, and reinforcing this state through IL-15–mediated persistence and suppression resistance, NK cells can be transformed into durable, high-function antitumor effectors.

This approach reframes immunotherapy as an exercise in **intracellular signal engineering**, offering a pathway toward generalized, antigen-independent cancer control.

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